

**PRAKTIKUM SISTEM OPERASI**

**MODUL 4**

**MEMBACA SOURCE CODE XINU**



Oleh:

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**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK**

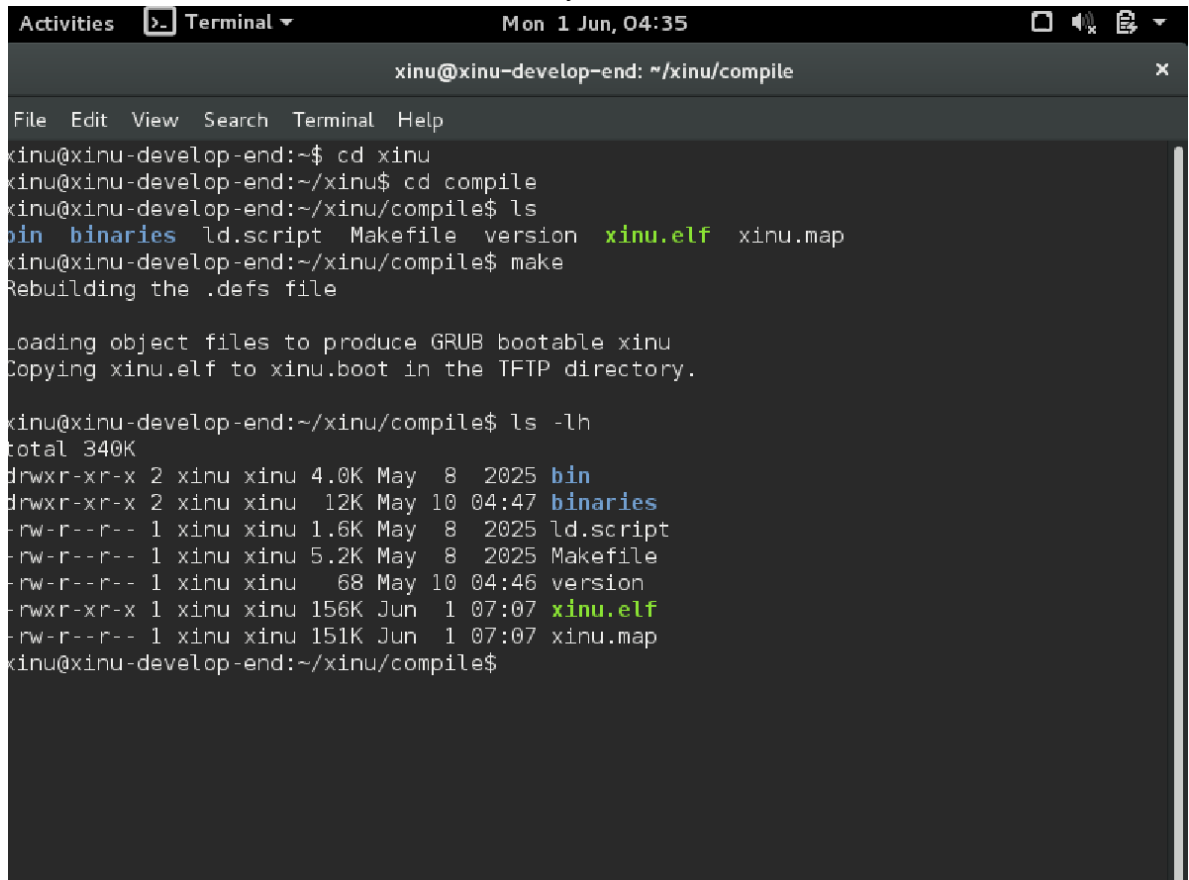
**DIREKTORAT KAMPUS PURWOKERTO**

**UNIVERSITAS TELKOM**

**2026**

## JURNAL

1. [10 Poin] Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?  
Hint: baca kembali modul-modul sebelumnya



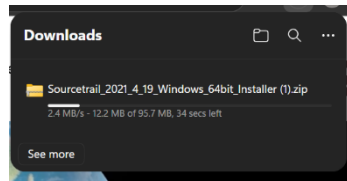
```
Activities Terminal Mon 1 Jun, 04:35
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ ls
bin binaries ld.script Makefile version xinu.elf xinu.map
xinu@xinu-develop-end:~/xinu/compile$ make
Rebuilding the .defs file

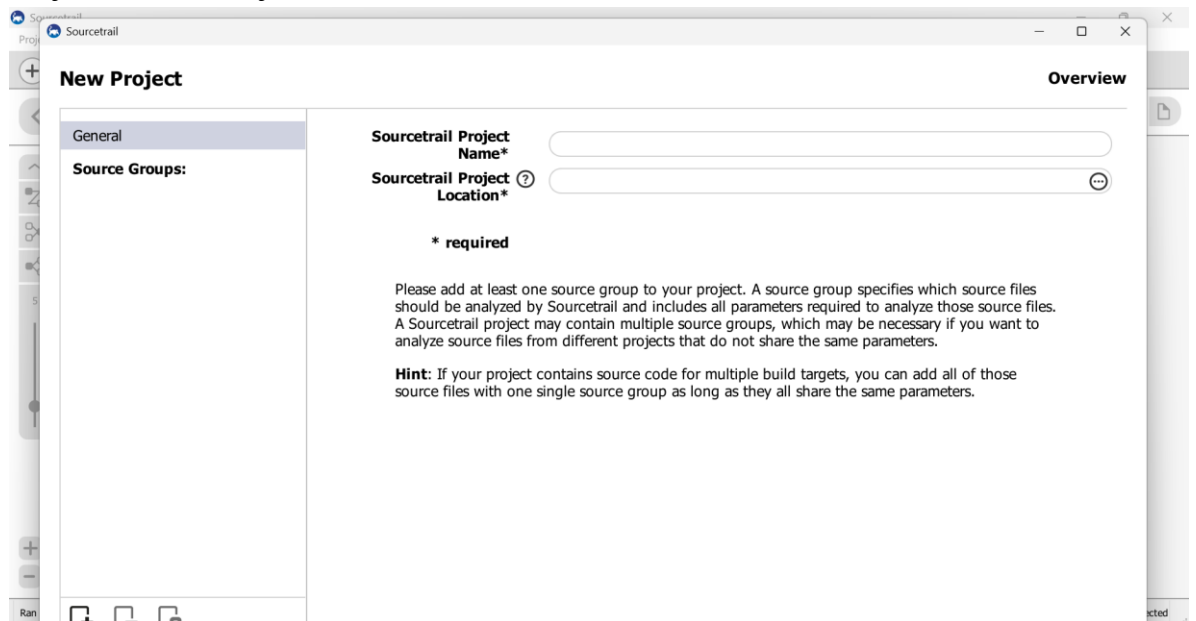
Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$ ls -lh
total 340K
drwxr-xr-x 2 xinu xinu 4.0K May  8  2025 bin
drwxr-xr-x 2 xinu xinu 12K May 10 04:47 binaries
-rw-r--r-- 1 xinu xinu 1.6K May  8  2025 ld.script
-rw-r--r-- 1 xinu xinu 5.2K May  8  2025 Makefile
-rw-r--r-- 1 xinu xinu  68 May 10 04:46 version
-rwxr-xr-x 1 xinu xinu 156K Jun  1 07:07 xinu.elf
-rw-r--r-- 1 xinu xinu 151K Jun  1 07:07 xinu.map
xinu@xinu-develop-end:~/xinu/compile$
```

- Nama image: xinu.elf
  - Ukuran file: 156k
  - Lokasi file: ~/xinu/compile
2. Membaca source code Xinu
    - a. Cek aplikasi bernama Sourcetrail di PC. Jika belum ada, download SourceTrail pada (DOWNLOAD SOURCETRAIL). SourceTrail adalah software untuk mengeksplorasi source code. Programmer yang bagus lebih banyak membaca kode daripada menulis kode



- b. Download file source code xinu yang tersedia pada attempt jurnal praktikum di LMS
- c. Jalankan SourceTrail
- d. Project → New Project



- e. Isi nama project xinu dan pilih lokasi project di manapun

Sourcetrail

## New Project

Overview

General

Source Groups:

Sourcetrail Project Name\* xinu

Sourcetrail Project Location\* C:/Semester 6/Praktikum SO/Arsip Praktikum 4

\* required

Please add at least one source group to your project. A source group specifies which source files should be analyzed by Sourcetrail and includes all parameters required to analyze those source files. A Sourcetrail project may contain multiple source groups, which may be necessary if you want to analyze source files from different projects that do not share the same parameters.

**Hint:** If your project contains source code for multiple build targets, you can add all of those source files with one single source group as long as they all share the same parameters.

+ -

Cancel Add Source Group Create

f. Add Source Groups, pilih C, lalu pilih Empty C Source Group

## NEW SOURCE GROUP

Type Selection

Source Group Types - C

C++ C Java Python Custom

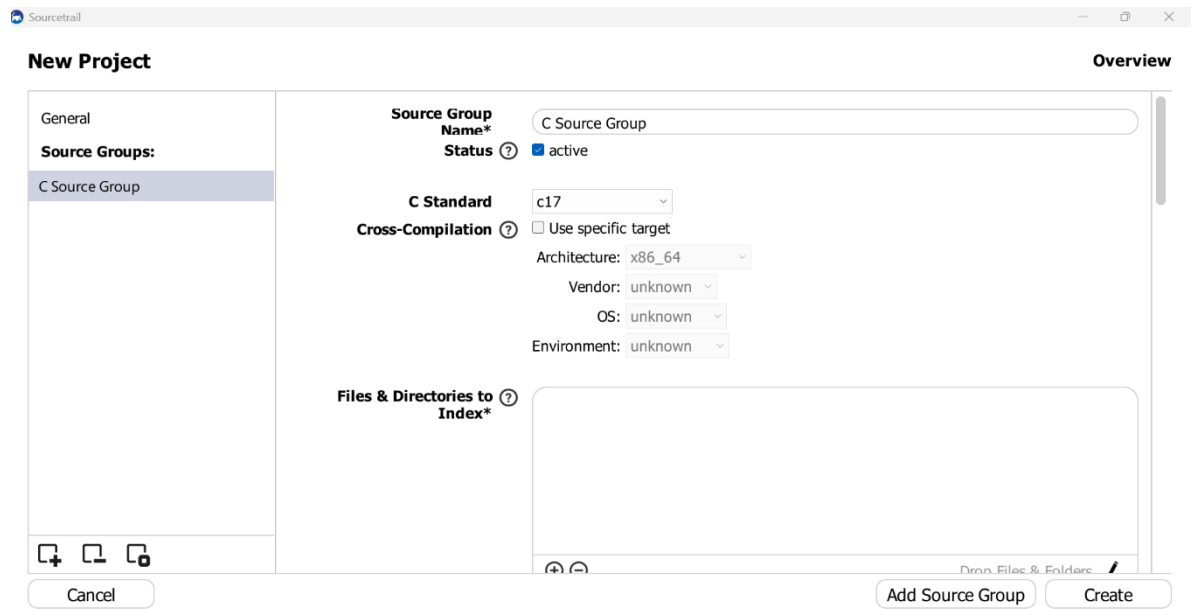
CDB VS CBP

C/C++ from Compilation Database\* C/C++ from Visual Studio C/C++ from Code::Blocks

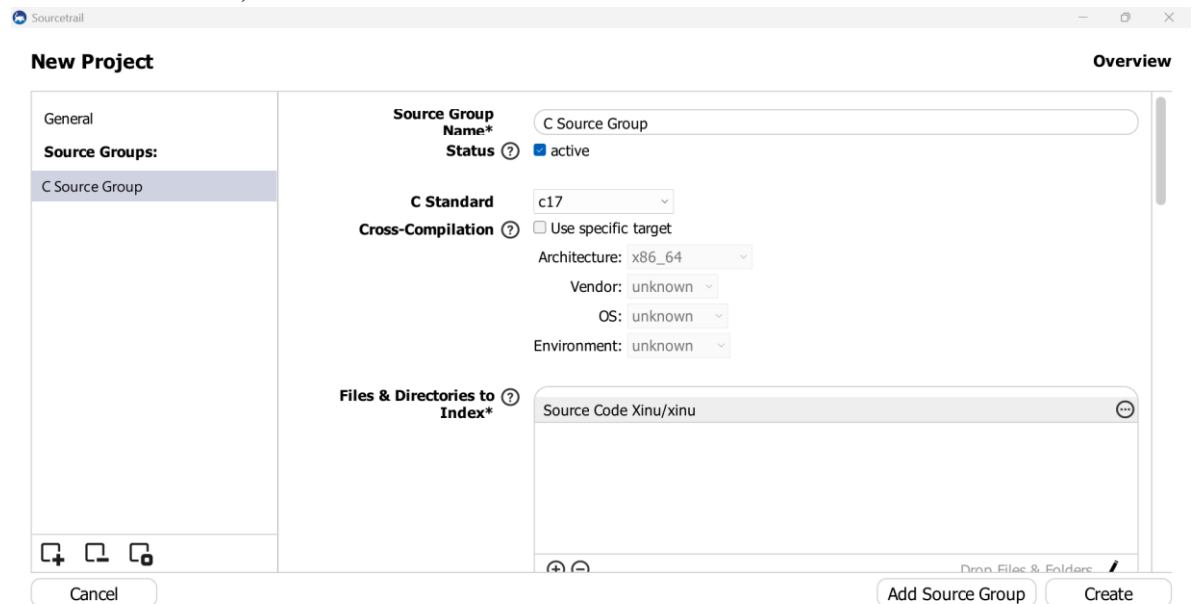
Empty C Source Group

Create a new Source Group by defining which C files will be indexed.

Cancel Previous Next



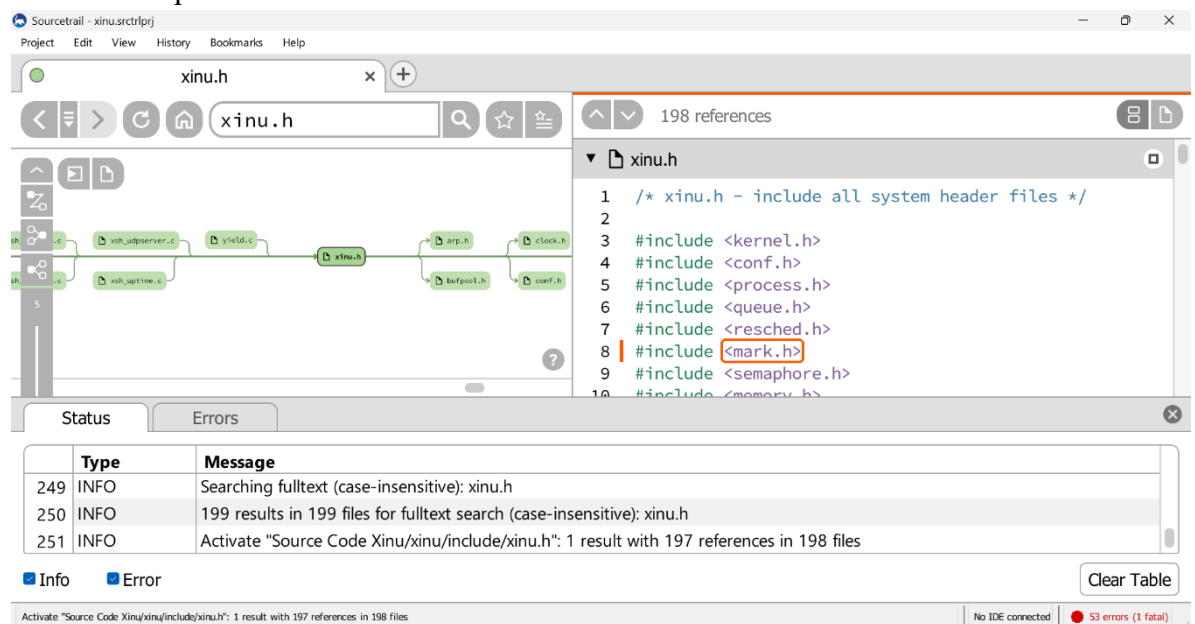
- g. File & Directories to Index: masukkan semua folder Xinu (yang sebelumnya telah di download)



- h. Include Paths: .../xinu/include



- i. Create
- j. Silahkan eksplorasi source code Xinu



Source trail - xinu.srctfprj
Project Edit View History Bookmarks Help

process.h

process.h

2 references

process.h

```

1  /* process.h - isbadpid */
2
3  /* Maximum number of processes in the system */
4
5  #ifndef NPROC
6  #define NPROC      8
7  #endif
8
9  /* Process state constants */

```

Status

Errors

|     | Type | Message                                                                                    |
|-----|------|--------------------------------------------------------------------------------------------|
| 250 | INFO | 199 results in 199 files for fulltext search (case-insensitive): xinu.h                    |
| 251 | INFO | Activate "Source Code Xinu/xinu/include/xinu.h": 1 result with 197 references in 198 files |
| 252 | INFO | Activate "Source Code Xinu/xinu/include/process.h": 1 result with 1 reference in 2 files   |

☒ Info
☒ Error

Clear Table

Activate "Source Code Xinu/xinu/include/process.h": 1 result with 1 reference in 2 files

No IDE connected

53 errors (1 fatal)

Source trail - xinu.srctfprj
Project Edit View History Bookmarks Help

initialize.c

initialize.c

1 references

initialize.c

```

1  /* initialize.c - nulluser, sysinit */
2
3  /* Handle system initialization and become the null process */
4
5  #include <xinu.h>
6  #include <string.h>
7
8  extern void start(void); /* Start of Xinu code */
9  extern void * end; /* End of Xinu code */

```

Status

Errors

|     | Type | Message                                                                                    |
|-----|------|--------------------------------------------------------------------------------------------|
| 256 | INFO | Searching fulltext (case-insensitive): initialize.h                                        |
| 257 | INFO | 0 results in 0 files for fulltext search (case-insensitive): initialize.h                  |
| 258 | INFO | Activate "Source Code Xinu/xinu/system/initialize.c": 1 result with 0 references in 1 file |

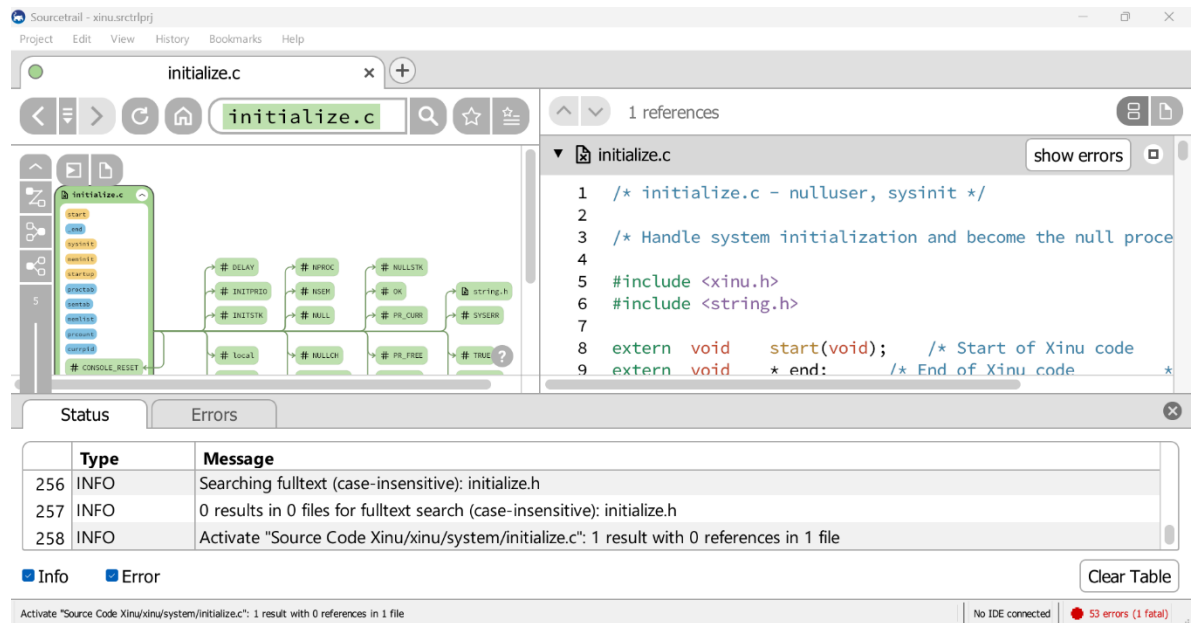
☒ Info
☒ Error

Clear Table

Activate "Source Code Xinu/xinu/system/initialize.c": 1 result with 0 references in 1 file

No IDE connected

53 errors (1 fatal)

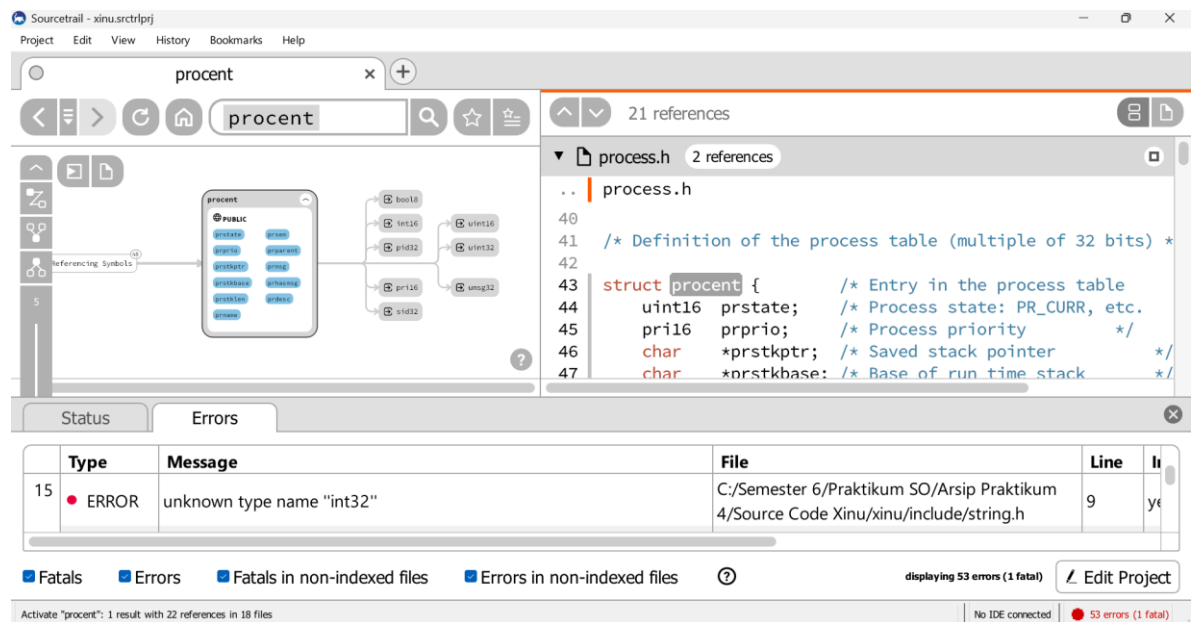


3. [10 Poin] Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut?  
Hint: file berektensi .h

Pada Linux bisa digunakan perintah “grep”

Contoh:

\$ grep -r “kata\_yang\_ingin\_dicari” /path/ke/direktori/tempat/file/berada





Struktur data proses pada sistem operasi Xinu disimpan pada file include/process.h dalam struktur struct procent. Struktur ini digunakan untuk menyimpan seluruh informasi yang berkaitan dengan proses seperti status proses, prioritas, stack, nama proses, semaphore, parent process, dan message yang dimiliki proses.

4. [80 poin] Mengubah welcome banner pada Xinu
    - a. Carilah file yang menyimpan banner Xinu!
- Hint: file berekstensi .h pada direktori xinu/include

```

/* shell.h - Declarations and constants used by the Xinu shell */

/* Size constants */

#define SHELL_BUFLen    TY_IBUFLen+1    /* Length of input buffer      */
#define SHELL_MAXTOK    32              /* Maximum tokens per line    */
#define SHELL_CMDSTK    8192            /* Size of stack for process  */
/*      that executes command    */
#define SHELL_ARGLEN    (SHELL_BUFLen+SHELL_MAXTOK) /* Argument area */
#define SHELL_CMDPRIO    20             /* Process priority for command */

/* Message constants */

/* Shell banner (assumes VT100) */

#define SHELL_BAN0      "\033[31;1m"
#define SHELL_BAN1      "-----"
#define SHELL_BAN2      " "
#define SHELL_BAN3      " "
#define SHELL_BAN4      " "
#define SHELL_BAN5      " "
#define SHELL_BAN6      " "
#define SHELL_BAN7      " "
#define SHELL_BAN8      "-----"
#define SHELL_BAN9      "\033[0m\n"

/* Messages shell displays for user */

```

File yang menyimpan banner yaitu shell.h

- b. Carilah file yang menampilkan banner Xinu!
- Hint: file berekstensi .c pada direktori xinu/shell

```

/* Print shell banner and startup message */
fprintf(dev, "\n\n%s\n%s\n%s\n%s\n%s\n%s\n%s\n%s\n%s\n",
        SHELL_BAN0, SHELL_BAN1, SHELL_BAN2, SHELL_BAN3, SHELL_BAN4,
        SHELL_BAN5, SHELL_BAN6, SHELL_BAN7, SHELL_BAN8, SHELL_BAN9);

fprintf(dev, "%s\n\n", SHELL_STRMSG);

/* Continually prompt the user, read input, and execute command */
while (TRUE) {
    /* Display prompt */
    fprintf(dev, SHELL_PROMPT);

    /* Read a command */
    len = read(dev, buf, sizeof(buf));

    /* Exit gracefully on end-of-file */
    if (len == EOF) {
        break;
    }

    /* If line only contains 'exit', exit gracefully */

```

File yang menampilkan banner yaitu shell.c

- c. Ubahlah welcome banner Xinu sehingga menjadi seperti ini:  
 Tambahkan NAMA dan NIM masing-masing, kemudian ubah banner Xinu  
 sesuai dengan selera Anda

```

/* Message constants */

/* Shell banner (assumes VT100) */

#define SHELL_BAN0 "\033[31;1m"
#define SHELL_BAN1 "-----Berlian Seva Astryana-----"
#define SHELL_BAN2 " "
#define SHELL_BAN3 " "
#define SHELL_BAN4 " "
#define SHELL_BAN5 " "
#define SHELL_BAN6 " "
#define SHELL_BAN7 " "
#define SHELL_BAN8 "-----2311104067-----"
#define SHELL_BAN9 "\033[0m\n"

/* Messages shell displays for user */

#define SHELL_PROMPT "xsh $ " /* Command prompt */
#define SHELL_STRMSG "Welcome to Xinu!\n" /* Welcome message */
#define SHELL_EXITMSG "Shell closed\n" /* Shell exit message */
#define SHELL_SYNERMSG "Syntax error\n" /* Syntax error message */
#define SHELL_CREATMSG "Cannot create process\n" /* command error */
#define SHELL_INERRMSG "Cannot open file %s for input\n" /* Input err */
#define SHELL_OUTERRMSG "Cannot open file %s for output\n" /* Output err */
#define SHELL_BGERRMSG "Cannot redirect I/O or background a builtin\n" /* Builtin cmd error message */
#define SHELL_PIPEMSG "Cannot create a pipe\n" /* error opening a pipe */

```

- d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux

- \$ cd xinu/compile
- \$ make clean
- \$ make
- \$ sudo minicom
- Jalankan vm backend

```
xinu@xinu-develop-end:~/xinu/system$ cd ..
xinu@xinu-develop-end:~/xinu$ cd include
xinu@xinu-develop-end:~/xinu/include$ gedit shell.h
xinu@xinu-develop-end:~/xinu/include$ cd ..
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make clean
```

```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
[0x0015E320 to 0x0009FFF7]
[0x00100000 to 0x005FBFFF]
116009 bytes of Xinu code.
[0x00100000 to 0x0011C528]
148424 bytes of data.
[0x0011FF00 to 0x001442C7]
Obtained IP address 10.0.3.15 (0x0a00030f)

-----Berlian Seva Astryana-----
XINU
-----2311104067-----

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:1 | ttyS0
```